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IV – CCSAD

ELECTIVE 4

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Assignment #3: Introduction to GitHub

Task 1: Introduction to Git and GitHub

Explain what Git is and how it differs from GitHub.

- Git is a tool that runs on computer. It tracks changes to files over time. Use it through commands on machine, and it works without an internet connection. GitHub is a website that stores Git projects online or called repositories. It let us back up our files, share code, and work with other people. Git is the tool on machine. GitHub is the cloud service that hosts our work repo.

Describe the purpose of version control in software development.

- The purpose of version control in software development is to avoid redundant of making a copies of file project like “project version 1” next is “version2” of one file that actually creating it avoid file clutter. Also if we make mistake we can jump back from the old version if something wrong happen so we can check it overtime and last the most important too is prevent conflicts like multiple developers can edit the same files at the same time without wiping out each other's work and these are branches, pull request, and merging.

Briefly explain the basic concepts of Git (e.g., repositories, commits, branches, pull requests, and merging)

- **Repository (Repo):** The main project folder. It holds files and the full log of every change we make.

Commit: A saved snapshot of the project. Each commit comes with a short message describing what changed.

Branch: A separate workspace split off from the main project. Using branches to test new features or fix bugs safely without touching working code.

Pull Request (PR): A request on GitHub to review and combine changes into the main project. Teammates read the code and leave feedback before accepting it.

Merging: The act of taking code from one branch and joining it into another.